Historical Transportability of Private-Capital SDFs

Which historical cash flows are relevant for pricing current funds?

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Abstract

Private-capital SDF estimation faces a three-way data constraint. There are few vintage years, older vintages may transport poorly to current funds, and the most recent vintages have not yet produced the realized cash flows needed for appraisal-free estimation. We propose a relevance-weighted hybrid estimator that treats the historical information set itself as a model-selection object. Seasoned vintages contribute realized-cash-flow NPV moments; unseasoned vintages can contribute short-horizon NAV Euler errors; and historical vintages are weighted by temporal proximity and similarity in credit spreads, interest rates, and public-market valuations. Expanding, rolling, decaying, and state-conditioned SDFs are compared on a common NAV-free criterion. An ex-post transportability matrix measures how well an SDF estimated in one historical window prices earlier and later realized vintages. A forward pseudo-real-time exercise then jointly selects the SDF specification, the historical weighting scheme, and whether unseasoned Euler moments improve subsequent realized-cash-flow pricing relative to excluding young funds. The paper does not claim that thirty vintage years can cleanly distinguish a stable law from every form of temporal instability. It asks the empirically sharper question: which use of the available history travels best out of sample?

1 Introduction

How much historical private-capital data should be used to estimate an SDF for a fund raised today? Under a stable and sufficiently mixing data-generating process, older vintages add information and a full-history estimator becomes increasingly precise. Under temporal instability, however, the same estimator may converge precisely to a historical mixture that is no longer the relevant pricing law for current funds. More funds within a vintage can reveal that vintage's pricing moment more accurately, but they cannot create additional independent economic histories.

This creates a three-way empirical constraint. First, the time-series dimension is intrinsically thin: even a long private-capital database contains only a few dozen vintage years, and adjacent vintages live through overlapping calendar histories. Second, increasing the sample by reaching further into the past requires a transport assumption: that pricing relations estimated on older funds remain relevant for funds raised under different financing, valuation, and exit conditions. Third, restricting estimation to fully realized cash flows removes precisely the recent vintages that are most informative about the current environment. Conventional resampling can reduce fund noise inside the observed sample, but it cannot create economic time or resolve this maturity delay.

The private-equity pricing literature has developed several ways to recover risk and performance from irregular, non-traded cash flows. Driessen et al. [2012] estimate factor exposures and abnormal performance directly from fund cash flows — the discounting framework our estimator adopts throughout — while Ang et al. [2018] recover a time series of private-equity returns from limited-partner cash flows. Korteweg and Nagel [2016] use a generalized public-market-equivalent approach to risk-adjust venture-capital cash flows. More recent work refines cash-flow benchmarking and SDF-based performance measurement [Gredil et al., 2023, Korteweg and Nagel, 2025]. These approaches differ in their maintained pricing restrictions and treatment of illiquidity, but they largely take the historical estimation sample as given. Our paper instead asks which part of that sample is informative for a current pricing problem: how well does an estimated private-capital SDF transport across economic time?

We call the empirical ability of an SDF estimated in one historical window to price cash flows from another window *historical transportability*. A stable pricing relation should travel across windows; a temporally unstable one should exhibit loss that changes with the distance and economic difference between estimation and evaluation periods. The concept does not require a literal regime classification. Smooth drift, isolated breaks, changes in the support of economic states, and changes in fund composition can all appear as failures of transportability. In an environment this short, a global test of "stability versus instability" is unlikely to have useful power against the many plausible alternatives. We therefore follow the forecasting-under-instability logic of evaluating local predictive performance rather than making a structural-break test the paper's central result [Pesaran and Timmermann, 2007, Rossi, 2021].

Our proposed solution is a *forward-selected relevance-weighted hybrid SDF*. It has three components. First, candidate SDFs are estimated under a small, pre-specified sequence of historical weighting schemes: the full expanding sample, hard rolling windows, exponential time decay, and a time-and-state kernel. Second, recent economic similarity is measured with three observable state variables fixed before evaluating fund performance: credit spreads, the interest-rate level, and public-market valuations. Third, recent unseasoned vintages enter through short-horizon NAV Euler restrictions, but their inclusion is not maintained by assumption: the hybrid estimator is compared with an otherwise identical estimator that excludes them. Every choice is judged on later realized cash flows.

The empirical design has two parts. An ex-post transportability matrix uses fully realized vintages to show how pricing laws travel forward and backward across historical windows. Its backward entries are structural diagnostics, not feasible forecasts. A pseudo-real-time exercise then recreates historical information dates, estimates each candidate using the information available at that origin, and scores it on cash flows realized later by the then-young funds. This second exercise provides the directional evidence required to choose a current estimator.

The paper makes four contributions.

1. It formulates the private-capital data problem as a joint constraint of few economic histories, potentially decaying historical relevance, and delayed cash-flow outcomes.
2. It introduces historical transportability matrices and signed-distance loss curves that show where and how far a private-capital SDF travels.
3. It proposes a relevance-weighted SDF that combines simple temporal windows with low-dimensional economic-state similarity, while enforcing an effective-sample-size and identification constraint.
4. It evaluates the informational value of unseasoned NAV Euler moments on a common, later-realized cash-flow target rather than on NAV fit.

2 Methodology

2.1 The Private-Capital Data Trilemma

The constraint introduced above has three connected parts: *few vintages*, because a few dozen vintage years with overlapping fund lives carry an effective temporal dimension smaller than the vintage count; *historical relevance*, because adding older vintages reduces sampling variance only if their pricing relations transport to the target period — under instability a precisely estimated full-history SDF can be precisely wrong for a current fund; and *delayed outcomes*, because the newest vintages contain the most current economic information but have not yet revealed their realized cash flows.

These are not three independent nuisances. Shortening the estimation window raises temporal relevance but increases sampling uncertainty and concentrates the sample in unseasoned funds. Requiring realized cash flows pushes the effective window backward. Reaching further into history restores moment count at the cost of a stronger transport assumption. The empirical objective is therefore not to eliminate any one component in isolation, but to find the combination that produces the lowest forward realized-cash-flow loss.

2.1.1 What the data can and cannot establish

Thirty vintage years do not support a clean omnibus decision between a time-invariant SDF and every form of temporal instability: a failure to reject stability may reflect low power, and a local episode of poor performance need not identify a permanent break. The paper therefore reports economically interpretable loss differences, transport curves, and model sets across a small pre-specified design space, treating local forecast performance as the target and parameter instability as one possible explanation for its variation [Rossi, 2021].

2.1.2 The cost of ignoring recent vintages

Discarding unseasoned vintages is clean only in a narrow measurement sense. It also changes the estimand by restricting the current SDF to funds raised far enough in the past to have substantially realized. Because buyout persistence and the composition of the industry have changed over time, this restriction cannot automatically be treated as innocuous [Harris et al., 2023]. The exclusion estimator is therefore an essential benchmark, not the default solution. Whether short-horizon Euler information improves forward pricing is an empirical question.

2.2 Pricing Moments

A candidate SDF specification M fixes a parametric family of one-period realizations $M_{t-1,t}(\theta)$ with $\theta \in \Theta_M$; dependence on M is left implicit whenever it is clear from context. Every specification in this paper belongs to the benchmark-discounting family of Driessen et al. [2012]: the one-period discount factor is the inverse gross return of a levered public benchmark,

$$M_{t-1,t}(\beta) = \frac{1}{1 + r_{f,t} + \beta\,(R_{m,t} - r_{f,t})},$$

so that $\theta = \beta$, with the multifactor case replacing the single excess return by a vector of factor excess returns. Setting $\beta = 0$ prices the risk-free asset and $\beta = 1$ a self-financing public-market investment exactly, period by period: benchmark pricing is built into the functional form rather than imposed through auxiliary moments, and no separate price-of-risk parameter exists. The abnormal-return intercept of the original method is fixed at zero; Section 2.3.1 states the resulting estimand.[1] Let $\Psi_{0,t}(\theta) = \prod_{s=1}^{t} M_{s-1,s}(\theta)$ denote the realized discount factor from fund inception to quarter t, with $\Psi_{0,0} = 1$, so that the time-zero capital call enters undiscounted by construction.

[1] Θ_M bounds β so that the levered benchmark return exceeds -100% in every sample quarter, keeping all discount factors positive.

2.2.1 Seasoned vintages

For seasoned fund i of vintage v with committed capital K_i, net dated cash flows $CF_{t,i}$ (calls negative, distributions positive), and last observed cash flow at T_i, the realized pricing error per committed dollar is

$$g_i^S(\theta) = \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta)\, CF_{t,i},$$

and the seasoned vintage moment is the equal-weighted fund mean $\epsilon_v^S(\theta) = N_v^{-1} \sum_{i \in v} g_i^S(\theta)$, where N_v is the number of admissible funds in vintage v; commitment-weighted vintage means are a pre-specified robustness variant.

The same object defines the paper's common evaluation currency. The realized evaluation loss of a parameter θ on a seasoned vintage v is

$$\widehat{U}_v^{\mathrm{real}}(\theta) = \epsilon_v^S(\theta)^2,$$

the squared vintage-mean realized pricing error. Squaring the vintage mean, rather than averaging squared fund-level errors, targets systematic mispricing and prevents idiosyncratic fund noise — which no SDF can price — from dominating design comparisons; absolute rather than squared vintage errors are a robustness check. Because $\Psi_{0,t}$ compounds many quarterly SDF realizations, a small number of extreme discount-factor paths can dominate both estimation and evaluation. We therefore report the concentration of loss contributions across vintages and a winsorized-loss robustness variant.

2.2.2 Unseasoned vintages

For an unseasoned fund, the full NPV moment is unavailable without valuing the residual claim. The baseline bridge uses the short-horizon NAV Euler error

$$e_{t,i}^H(\theta) = M_{t-1,t}(\theta) R_{t,i}^H - 1,$$

where $R_{t,i}^H$ is the quarterly Horizon-IRR return accounting for interim calls and distributions. The unseasoned vintage moment scales the pooled fund-quarter mean into the units of the seasoned block:

$$\epsilon_v^U(\theta) = \bar{D} \cdot \frac{1}{n_v^U} \sum_{(i,t) \in v} e_{t,i}^H(\theta),$$

where n_v^U is the number of admissible fund-quarters of vintage v and $\bar{D}$ is the median discounted duration, in quarters, of seasoned funds in the estimation sample, fixed before any out-of-sample comparison. The scaling has a first-order justification: a pricing error of e per quarter sustained over a fund's life accumulates to a lifetime NPV error of approximately duration times e, so both blocks are expressed in NPV per committed dollar and can legitimately share one set of relevance weights. Euler errors are never discounted back to inception and never enter the comparison score. Reported NAVs are noisy appraisals, so this block is treated as a candidate information bridge rather than a traded-return restriction that must hold exactly. State-space corrected NAVs are a robustness extension, not the baseline [Brown et al., 2023].

The main unseasoned-treatment comparison is deliberately parsimonious:

1. *Exclude*: estimate only with seasoned realized-cash-flow moments;
2. *Hybrid*: add short-horizon NAV Euler moments for unseasoned vintages.

Both are evaluated on the identical subsequently realized cash flows. A more complicated measurement model earns a place in the main design only if the simple comparison demonstrates economically meaningful value from recent NAV information.

2.3 The Relevance-Weighted Hybrid SDF

At target date T, let z_v collect the economic state associated with vintage v. The baseline state vector contains three pre-performance variables: a credit spread, the interest-rate level, and a public-market valuation ratio. The variables are measured at or immediately before vintage formation and standardized using the estimation sample only.

For historical vintage v, define the unnormalized relevance weight

$$w_{v,T}(h,b) = \exp\left[-\log(2)\frac{T-v}{h}\right]\exp\left[-\frac{1}{2b^2}(z_v - z_T)'(z_v - z_T)\right],$$

where h is a time half-life and b is a common state-similarity bandwidth. The simple common bandwidth avoids estimating three separate state-sensitivity parameters from a few dozen vintages. Setting $h = \infty$ removes time decay; setting $b = \infty$ removes state conditioning. Hard rolling windows are obtained by replacing the time-decay component with an indicator.

Let q_v be a reliability weight, set to the vintage fund count N_v in the baseline; vintage-equal weighting ($q_v = 1$) is a pre-specified robustness variant. The products $w_{v,T}q_v$ are normalized to sum to one jointly across all vintages entering the objective, seasoned and unseasoned alike, so that a vintage's relevance does not depend on the block through which it enters. For candidate unseasoned treatment $A \in \{\text{Exclude}, \text{Hybrid}\}$, estimate

$$\widehat{\theta}_T(M,h,b,A) = \arg\min_{\theta\in\Theta_M}\left\{\sum_{v\in\mathcal{V}_T^S}\widetilde{w}_{v,T}\epsilon_v^S(\theta)^2 + \mathbf{1}\{A = \text{Hybrid}\}\sum_{v\in\mathcal{V}_T^U}\widetilde{w}_{v,T}\epsilon_v^U(\theta)^2\right\}.$$

We collect a complete design in the tuple $c = (M,h,b,A)$ and write $\widehat{\theta}_T(c)$ below. Because unseasoned vintages are by construction the most recent, time decay concentrates weight on the Euler block: under short half-lives the hybrid objective becomes increasingly NAV-driven. Every configuration therefore reports its realized unseasoned weight share $\pi_U(T;h,b) = \sum_{v\in\mathcal{V}_T^U}\widetilde{w}_{v,T}$, which makes this interaction visible rather than leaving it implicit in the kernel.

The design sequence moves from transparent to reasonably sophisticated while remaining feasible in a thin vintage panel:

1. **Expanding**: equal weight on the full available history;
2. **Rolling**: equal weight inside a fixed recent-vintage window;
3. **Decay**: exponential time weights indexed by half-life h;
4. **Time–state**: exponential decay combined with similarity in credit spreads, interest rates, and public valuations.

A regularized time-varying parameter path of the form

$$\min_{\{\theta_v\}}\sum_v \epsilon_v(\theta_v)^2 + \kappa\sum_{v>1}\|\theta_v - \theta_{v-1}\|_2^2$$

is the principal sophistication check [Cui et al., 2024]. Its target-date parameter is the final element of the estimated path, and κ is selected on strictly earlier origins under the sequential protocol of Section 2.7. It is not the headline estimator unless it produces a clear forward improvement over the lower-dimensional weighting schemes.

Every candidate must satisfy two feasibility restrictions before it can be scored: the weighted moment Jacobian must be sufficiently well conditioned, and the weight concentration must leave an effective vintage count

$$V_{\text{eff}}(T) = \frac{1}{\sum_v \widetilde{w}_{v,T}^2}$$

comfortably above the number of SDF parameters. The count is computed both over all vintages and, with weights renormalized within the block, over the seasoned block alone; the threshold must hold for the seasoned block, since realized cash flows anchor identification. This prevents an apparently local estimator from being driven by one historically similar vintage or propped up by appraisal-based moments alone.

2.3.1 Pricing law versus performance measurement

The estimator is a pricing-law estimator, not a performance measure. With the abnormal-return intercept fixed at zero, average in-window performance is absorbed into the exposure estimate through the premium channel: to first order the fitted exposure is $\beta^* \approx \beta + \alpha/\lambda$, where α is average abnormal performance and λ the factor premium. The estimand is therefore the composite pricing-law beta — the leverage on the public benchmark that best rationalizes the weighted history — not a pure risk exposure. Freeing the intercept would separate the two in principle, but a per-period intercept compounded over decade-long fund lives is fragile in a thin vintage panel, and we exclude it by design. Statements about risk-adjusted performance in this paper therefore refer exclusively to out-of-window residuals — realized pricing errors of vintages that did not enter, or received negligible weight in, the estimation sample. In-window residuals are diagnostics of fit, never evidence of abnormal performance. Section 2.4.3 turns this absorption into a testable implication for out-of-window residuals.

2.4 Historical Transportability

2.4.1 The transportability matrix

Let s index a historical estimation window and t an excluded, fully realized evaluation vintage. Throughout this subsection s denotes the window's information end date; where present, the time-decay and state kernels are centered on s and z_s, and $A = \text{Exclude}$ is imposed. Define the NAV-free transport loss

$$L_{s\to t}(c) = \widehat{U}_t^{\text{real}}\left(\widehat{\theta}_s(c)\right),$$

where the evaluation score uses only realized fund cash flows. The exclusion version isolates historical transportability from the separate question of how young-fund NAV information enters. Stacking these losses over estimation windows and evaluation vintages yields a *historical transportability matrix*. Its rows are window SDFs, its columns are evaluation vintages, and each entry measures how well a pricing law travels across economic time.

Aggregating by the signed temporal distance $d = t - s$, measured from the window end, gives the transport loss curve

$$\mathcal{T}_c(d) = \frac{1}{N_d} \sum_s L_{s\to s+d}(c).$$

Positive distances measure forward transportability; negative distances measure backward transportability. Only the former has a real-time forecasting interpretation. Backward transportability is a diagnostic of structural stability, not a feasible investment rule.

The matrix is descriptive by design. A flat surface is consistent with broad historical transportability; loss increasing in $|d|$ is consistent with temporal localization; and isolated high-loss columns indicate historically unusual target periods. None of these patterns alone proves a particular structural-break model.

2.4.2 Calendar-time overlap

Adjacent vintage moments share public-factor histories because their funds remain active for many years. We therefore measure pairwise calendar-time overlap rather than treating vintage labels as independent by construction. Let $a_{v,t}$ denote vintage v's absolute discounted cash-flow exposure to calendar period t, evaluated at reference parameter values fixed in advance.[2] Define

$$O_{vv'} = \frac{\sum_t a_{v,t} a_{v',t}}{\sqrt{\sum_t a_{v,t}^2}\sqrt{\sum_t a_{v',t}^2}}.$$

The overlap matrix is reported alongside the transportability matrix. Estimation–evaluation pairs with overlap above a pre-specified threshold are flagged, and headline comparisons are re-reported excluding them. Because complete separation would eliminate most of the data, overlap is quantified and stress-tested rather than treated as a binary admissibility standard.

2.4.3 Absorbed premia, omitted factors, and the meaning of transport loss

The premium channel of Section 2.3.1 has a testable implication. Let $\widehat{\lambda}_s$ and $\widehat{\lambda}_t$ denote the realized benchmark premia over the weighted estimation window and over the cash-flow-weighted life of evaluation vintage t. If true performance is (α, β), estimation forces $\beta^* = \beta + \alpha/\widehat{\lambda}_s$, and the signed transport residual is, to first order and in per-period units,[3]

$$\epsilon_t^S\left(\widehat{\theta}_s\right) \approx \alpha\left(1 - \frac{\widehat{\lambda}_t}{\widehat{\lambda}_s}\right),$$

linear in the realized evaluation premium with intercept α and slope $-(\beta^* - \beta)$. Under the composite null $\alpha = 0$ the line is flat at zero. Regressing signed transport residuals on realized evaluation premia is therefore both a specification test of the zero-intercept design and, upon rejection, a first-order recovery of the split: the intercept estimates α and $\beta = \beta^* + \text{slope}$. With few effective windows this *premium-exposure diagnostic* is reported with uncertainty bands, as a diagnostic rather than a second estimator.

The same channel governs omitted factors. Decompose an omitted factor into its projection on the included benchmarks and an orthogonal remainder. The spanned component shifts the included exposures by the usual omitted-variable term and is harmless for pricing while factor correlations are stable. The unspanned premium component is observationally identical to α: skill and unspanned factor premia are the same object in cash-flow data and are absorbed together into β^*. Three corollaries follow.

1. **Constant unspanned premia transport.** A stable absorbed component enters β^* identically in every window; transport residuals remain mean-zero, and only their variance rises with premium variability. The composite prices without the true factor model.
2. **Time-varying unspanned premia are what transport failure detects.** A premium that shifts across regimes makes a window-fitted β^* systematically misprice other windows. Loss rising with temporal or

[2]A Jacobian-based exposure measure yields an alternative overlap matrix and is reported as a variant.

[3]NPV residuals per committed dollar scale with fund duration; dividing by $\bar{D}$ of Section 2.2 expresses them in the per-period units used here.

state distance therefore has a direct asset-pricing reading, and the premium-exposure diagnostic extends to candidate premia — realized credit or illiquidity-proxy returns — to name the missing factor.

3. **The time–state kernel is the reduced-form hedge.** Conditioning estimation windows on credit spreads, rates, and valuations estimates the composite locally in states where omitted premia are plausibly similar. The kernel does not claim to know the missing factors; it conditions on their observable proxies.

The multifactor menu addresses only the spanned component; the unspanned component is invisible inside any single window and visible only through transport. This is the information-theoretic limit of cash-flow data, and it is why the paper selects on forward transport loss rather than on in-window fit.

2.5 Unseasoned Vintages as Delayed Outcomes

At historical information date τ, estimate $\widehat{\theta}_\tau(c)$ by minimizing the weighted objective of the previous subsection using only fund information dated no later than τ. Candidate bridges are compared on the identical realized-cash-flow target after the then-unseasoned vintages have seasoned:

$$L_{\tau\to\tau+\ell}(c) = \frac{1}{|\mathcal{V}_\tau^U \cap \mathcal{V}_{\tau+\ell}^S|} \sum_{v\in\mathcal{V}_\tau^U\cap\mathcal{V}_{\tau+\ell}^S} \widehat{U}_v^{\mathrm{real}}\Big(\widehat{\theta}_\tau(c)\Big).$$

One convention deserves emphasis. The time and state kernels remain centered on the origin, τ and z_τ, while the scored vintages were raised shortly before τ under possibly different states. The forward loss therefore validates the estimator as a pricing rule for recently raised funds, and its application to a fund raised at the current date extrapolates the transport pattern by one further step. We state this openly rather than re-center the kernel on each evaluation vintage, which would replace a single origin SDF with a vintage-indexed family at substantial computational cost and little expected gain when origin and formation states are close.

The empirical comparison therefore selects a triplet: the SDF specification, the historical weighting rule, and the unseasoned-vintage treatment. Exclusion remains in the race as the benchmark measuring the cost of delayed outcomes; the short-horizon Euler block is the proposed simple bridge. The paper presents the configuration with the lowest forward loss as its preferred empirical solution, subject to the feasibility and stability diagnostics above.

2.6 Evaluation Design

Ex-post transportability. Estimate local SDFs on fully observed historical windows and score them on earlier and later seasoned vintages. This exercise is fully NAV-free and isolates temporal transportability, but it does not claim a real-time interpretation when later information is used to estimate an SDF that is scored on an earlier period.

Pseudo-real-time delayed-outcome validation. Reconstruct a sequence of historical information dates, estimate each window/bridge combination using the information fields available at the corresponding date, and score it on subsequently realized fund cash flows. A literal real-time reconstruction would also require historical database snapshots, contemporaneous coverage records, and knowledge of which observations were later backfilled. Imposing that ideal would leave essentially no usable evaluation sample. We therefore adopt the pragmatic pseudo-real-time standard feasible in the data and state its residual look-ahead limitation explicitly.[4]

[4]The fully reconstructed real-time ideal is conceptually useful even when empirically infeasible: it identifies the remaining source of look-ahead without pretending that an empty sample is a superior research design.

The design grid is intentionally coarse and common across SDF specifications; Section 2.7 separates the paired comparison of fixed designs from the sequential evaluation of the selection rule itself. A further structural limitation is timing: because a vintage needs roughly a decade to season, the last scoreable origin precedes the end of the sample by approximately one seasoning lag. Statements about funds raised at the current date therefore extrapolate a selection rule whose direct validation necessarily ends earlier, and the paper says so rather than implying that pseudo-real-time evidence reaches the present.

2.7 Forward Selection Criterion

Let O denote the admissible pseudo-real-time origins and $\mathcal{H}_\tau$ the vintages that were unseasoned at τ and later become eligible for realized-cash-flow scoring. For design $c = (M, h, b, A)$, define

$$\widehat{R}(c) = \frac{1}{|O|} \sum_{\tau \in O} \frac{1}{|\mathcal{H}_\tau|} \sum_{v \in \mathcal{H}_\tau} \widehat{U}_v^{\text{real}}\left(\widehat{\theta}_\tau(c)\right).$$

All designs face the same origins and realized evaluation funds, so comparisons are paired. Two protocols use this criterion for different purposes.

Protocol A: fixed-design comparison (headline). Every design in the pre-specified grid is estimated at every admissible origin and scored on the identical origin–vintage cells. No tuning occurs; the coarse grid is the pre-registration. The preferred design minimizes $\widehat{R}(c)$ and is reported together with the model confidence set of designs whose forward loss is not significantly worse [Hansen et al., 2011], so that small differences are not given a false winner-takes-all interpretation.

Protocol B: sequential selection (implementability check). Because the Protocol A minimizer is chosen using all origins, we also evaluate the selection rule itself: at each origin τ, the design minimizing $\widehat{R}$ computed over strictly earlier origins is scored on origin τ alone. The average of these sequentially selected losses measures what forward selection would have delivered in real time. With few origins this estimate is noisy, and it is reported as a check on Protocol A rather than as the headline comparison.

The paper emphasizes effect sizes, origin-by-origin stability, and sensitivity to the coarse grid. With only a few dozen vintage years, formal rejections of equal predictive loss are interpreted cautiously.

2.8 Inference and Reporting

Model comparisons use paired loss differences because every admissible design is evaluated on the same origin–vintage cells [Giacomini and White, 2006]. Losses are first averaged within origin so that origins with many funds or newly seasoned vintages do not dominate mechanically. Because adjacent annual origins share most of their evaluation vintages, the same realized outcome enters $\widehat{R}$ at several origins, and the effective number of independent evaluations is far smaller than $|O|$; the number of distinct evaluation vintages and funds behind each comparison is therefore reported alongside the origin-coverage table. We report the mean paired difference, its distribution across origins, and block-bootstrap uncertainty that preserves adjacent-origin dependence. The premium-exposure diagnostic of Section 2.4.3 is reported alongside these comparisons. Parameter uncertainty for a selected SDF is assessed with vintage- level resampling and calendar-overlap blocks. Given the short time series and the additional uncertainty created by model selection, confidence intervals and predictive-loss tests are interpreted as measures of precision rather than as a binary stability verdict.

3 Data and Empirical Design

This section maps the empirical inputs directly into the moments, state weights, and forward loss defined above. The ordering is deliberate: we first construct an appraisal-free realized-cash-flow target, then define what was observable at each historical origin, and only then compare alternative uses of the available history.

3.1 Private-Capital Cash-Flow Data

The fund data are drawn from **[TBD: data provider and extraction date]**. The unit of observation is a dated limited-partner cash flow for a **[TBD: buyout / broader private-capital]** fund. Required fields are fund identifier, vintage year, commitment, dated capital calls, dated distributions, reported NAVs, strategy, geography, and — where available — general-partner identifier. Cash flows and NAVs are measured net of management fees and carried interest from the limited partner's perspective and converted to **[TBD: currency and real/nominal units]**. We exclude funds with internally inconsistent cash-flow histories, missing commitments, nonpositive commitments, or implausible scale ratios under a pre-specified cleaning rule. The analysis retains all admissible funds within a vintage; it does not create pseudo-vintages by resampling funds.

The first data table should make the thin time-series dimension visible rather than reporting only a large fund count:

Table 1: Private-capital sample by vintage period

Vintage period	Vintages	Funds	Commitments	Funds/vintage	Realized share
[TBD: early period]					
[TBD: middle period]					
[TBD: recent period]					
Full sample					

Notes: Commitments are reported in **[TBD: units]**. Funds per vintage is the median across vintage years. Realized share is the fraction of funds satisfying the baseline seasoning rule at the final sample date.

3.2 Seasoning and the Realized Evaluation Sample

At information date τ, a fund is seasoned when its residual value share

$$s_{i,\tau} = \frac{\mathrm{RVPI}_{i,\tau}}{\mathrm{DPI}_{i,\tau} + \mathrm{RVPI}_{i,\tau}}$$

is at most δ. The proposed baseline is $\delta = 0.15$, with 0.10 and 0.20 as pre-specified robustness thresholds. A fund that fails this rule may contribute to the hybrid short-horizon Euler block but never to the realized evaluation loss at that date. The realized target includes only calls and distributions observed by the evaluation date; the residual NAV is used to determine eligibility but is neither priced nor added as a terminal cash flow. Thus, δ trades bounded terminal truncation for sample size without fitting the score to NAV; $\delta = 0$ is the strict robustness check.

We report, for every pseudo-real-time origin, the number of available vintages, seasoned and unseasoned funds, the fraction of commitment represented by the realized evaluation set, and the later vintages that can eventually be scored. This origin-by-origin coverage table is essential: it reveals whether an apparent forecasting gain comes from a genuinely broader information set or from a small and unusual set of funds that happens to mature early.

3.3 Public Factors and Candidate SDFs

Public-market returns and the risk-free rate are obtained from **[TBD: source]** at the same frequency as the cash-flow discounting exercise. All candidate SDFs are members of the single benchmark-discounting family of Section 2.2, and the main comparison uses a deliberately small nested set: $\beta = 0$ (risk-free discounting), $\beta = 1$ (the public-market-equivalent benchmark), a freely estimated one-factor β, and a freely estimated low-dimensional multifactor β vector. This keeps the empirical question focused on the historical information set rather than on kernel functional forms or a factor zoo. Factor definitions, currency, return convention, and the mapping from quarterly factors to dated cash flows are fixed before inspecting forward pricing losses.

3.4 Economic-State Variables

The state vector z_v contains one variable for each of three economic dimensions: a corporate credit spread, the interest-rate level, and a public- equity valuation ratio. The proposed baseline definitions are the high-yield option-adjusted spread, a long-term nominal government yield, and the cyclically adjusted price–earnings ratio **[TBD: or a price–dividend ratio]**. To avoid using fund outcomes to define similarity, each variable is measured as its average over the 12 months ending immediately before the vintage year. At each origin, levels are standardized using only the available estimation vintages. We also report the historical support of z_T; a target outside that support is an extrapolation problem and should not receive a falsely precise local estimate.

3.5 Pseudo-Real-Time Origins

Annual origins begin at the first date for which all candidate SDFs satisfy the minimum effective-vintage and Jacobian conditions. At each origin, cash flows, NAVs, factors, and state variables are truncated to their admissible information date. Under Protocol A no origin-level tuning occurs; under the sequential Protocol B, design selection at each origin uses strictly earlier origins only. The final extract does not reconstruct historical database coverage or later backfills; this pragmatic limitation is disclosed and tested by excluding observations with suspiciously late first reporting where such timestamps exist. The stronger literal real-time standard would otherwise reduce the usable evaluation sample to essentially zero.

3.6 Pre-Specified Design Grid

The grid is intentionally coarse. Its final endpoints are fixed after a descriptive feasibility audit, but before comparing out-of-sample losses.

3.7 Empirical Workflow

The empirical analysis proceeds in the same order as the methodology:

1. construct seasoned realized-cash-flow moments and document calendar-time overlap across vintages;

Table 2: Pre-specified empirical design

Object	Baseline candidates	Restriction or robustness
SDF (benchmark β)	$\beta = 0$; $\beta = 1$ (PME); estimated β; multifactor β	Common factor set at every origin
Historical weighting	Expanding; rolling; exponential decay; time–state	Same candidate sequence for every SDF
Rolling length	8, 12, 16 vintages	Retained only when identification is adequate
Time half-life	4, 8, 12 vintage years	$h = \infty$ reproduces no decay
State bandwidth	0.75, 1.5, 3 standardized units	$b = \infty$ reproduces time-only weighting
Young-fund treatment	Exclude; short-horizon NAV Euler hybrid	Common later realized-cash-flow target
Seasoning threshold	Residual value share at most 15%	10%, 20%, and zero residual value
Feasibility	$V_{\text{eff}} \geq \max\{8, 3\dim(\theta)\}$, incl. seasoned block	Jacobian conditioning reported separately

2. estimate fully realized local SDFs and form the ex-post historical transportability matrix;
3. reconstruct pseudo-real-time origins and calculate feasible expanding, rolling, decaying, and time–state estimates;
4. select tuning parameters using earlier origins only;
5. score all candidates on the same later realized-cash-flow outcomes and compare exclusion with the short-horizon NAV Euler hybrid; and
6. re-estimate the selected specification at the final information date and report both its current SDF parameters and the historical weights that support them.

4 Empirical Results

4.1 The Effective Historical Sample

Table 1 first documents the nominal and effective sample size. A vintage-by-calendar-year cash-flow heat map shows the long overlap of fund lives, while the overlap matrix O summarizes common factor exposure. We report the eigenvalue-based effective temporal rank and effective vintage counts under each candidate weighting rule. This section establishes how much independent economic history is actually available before asking whether old history remains relevant.

4.2 Ex-Post Historical Transportability

The first main figure is the historical transportability matrix for the baseline SDF. Rows identify estimation windows, columns identify excluded realized vintages, and colors represent realized pricing loss. Signed-distance loss curves then summarize forward and backward transport. The empirical questions are whether loss rises with temporal distance, whether the pattern is asymmetric, and whether particular evaluation vintages are difficult for every historical window. Signed residuals are also plotted against realized evaluation-window premia, implementing the premium-exposure diagnostic of Section 2.4.3.

4.3 How Much History Travels Forward?

The main model-comparison table reports paired forward loss for the expanding, rolling, decay, and time–state estimators on common origin–vintage cells. For each design we report average loss, the difference relative to expanding history, origin-level uncertainty, V_{eff}, and the fraction of origins passing identification checks. This table supplies the paper's primary answer: whether using less, or selectively weighted, history improves pricing of later realized private-capital cash flows.

4.4 Does Economic-State Similarity Add Information?

We next hold the time-weighting rule fixed and compare time-only with time–state weights. Weight plots show which distant vintages are recovered because their credit spreads, interest rates, or public valuations resemble the target. Leave-one-state-variable-out results reveal whether improvement is broad-based or driven by a single state proxy. A state-conditioned estimator is preferred only if it improves forward loss without collapsing the effective vintage count.

4.5 Do Unseasoned Vintages Improve the Current SDF?

For every admissible origin, we compare the exclusion and short-horizon NAV Euler estimators using identical later realized cash flows. The main table reports paired loss differences overall and by the age of the young funds at the estimation origin. We also relate the hybrid's gain to its NAV share and subsequent NAV revisions. This distinguishes useful early information from a mechanical fit to smoothed reported values.

4.6 The Preferred SDF and the Pricing of Recent Vintages

The final results section re-estimates the preferred design at the current information date. We report the estimated benchmark exposures, the weights assigned to every historical vintage, and the out-of-window pricing residuals of recent vintages under the estimated law. In-window residuals are shown only as fit diagnostics, since average in-window performance is absorbed by construction. Sensitivity is reported across all configurations in the model confidence set. The conventional expanding-history estimate is shown alongside it so that the effect of historical transportability is transparent.

5 Robustness and Method Validation

5.1 Alternative Sample and Moment Choices

Robustness exercises vary the seasoning threshold, exclude residual NAV entirely, remove the financial-crisis and pandemic vintages one at a time, alter fund-size reliability weights, and restrict the sample by strategy or geography where coverage permits. Results are also reported with vintage- equal weighting so that a few data-rich vintage years cannot determine the conclusion.

5.2 Alternative Windows, States, and SDFs

We vary the coarse window grid, use a real rather than nominal long rate, replace the valuation proxy, winsorize state distances, and compare the common bandwidth with a strongly regularized variable-specific bandwidth. The smooth time-varying parameter path is retained as a sophistication check. Alternative SDFs matter only insofar as the transportability result survives reasonable pricing specifications.

5.3 Placebos

Two placebos discipline the interpretation of state similarity. First, state vectors are permuted across vintages while their time ordering is retained. Second, the target state is replaced by a future state that was unavailable at the origin. Genuine transport gains should disappear under the first placebo; the second quantifies how much an inadvertently forward-looking state construction could exaggerate performance.

5.4 Compact Method Validation

A standalone simulation section is not required for the paper's main empirical claim. The estimator is selected by observed forward realized-cash-flow loss, and a large menu of simulated breaks would not establish which process generated the single historical private-capital sample. A compact appendix calibration can nevertheless provide a useful diagnostic. It should match the empirical vintage count, cash-flow-age profile, and calendar overlap under only two designs — a stable SDF and a smoothly drifting SDF — each with and without appraisal smoothing. Its limited purpose is to verify that the selection procedure does not mechanically prefer short windows, state conditioning, or the NAV bridge when the pricing law is stable. The empirical transport and delayed-outcome exercises remain the evidence of interest.

6 Conclusion

Private-capital SDF estimation cannot solve a thin and delayed economic history by resampling funds more aggressively. The relevant choice is which historical vintages to transport to the target period and whether partial information from recent funds improves that transport. The proposed estimator makes both choices empirical: it weights history by time and observable economic similarity, admits recent NAV information only through a short-horizon bridge, and selects the complete design by its ability to price later realized cash flows. The result is a current SDF supported by the part of history that has demonstrated the greatest forward relevance, rather than by the longest history available.

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